

Component 03 - UEF-Net: Uncertainty-Aware Ordinal EF Severity Grading

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Project R26-IT-083 | Component 03

Technical dossier compiled from the codebase

REVISION NOTE (post-audit remediation). This dossier was compiled, then the issues it identified were fixed in the codebase. Items marked **[FIXED]** below have been resolved; items marked **[CORRECTED]** were errors in the *first version of this dossier itself* and are retracted. §25 records the final state of every issue.

Author: Dilukshan Viyapury (IT22219534) · Project ID R26-IT-083, Component 03 **Repository root:**

c:\Users\diviya\Desktop\Component_03 **Dossier compiled:** from codebase state at commit e3b18e6

0. Component Abstract

Left-ventricular ejection fraction (EF) is graded manually from echocardiogram video, a process carrying 4–5 EF points of inter-observer variability (README.md §1). This component implements **UEF-Net**, an R(2+1)D-18 spatio-temporal network with four prediction heads (EF regression, ordered-cutpoint ordinal, auxiliary softmax class, log-variance) trained on EchoNet-Dynamic with harmonized CAMUS co-training, producing both continuous EF and a four-class severity grade under ~11:1 class imbalance. On the untouched 1,277-study test split a three-seed ensemble achieves **MAE 3.979 EF points, R² 0.818, 73.0 % overall accuracy, 73.7 % balanced accuracy, and 0.723 minimum per-class recall**, with 99.7 % of predictions within one severity class and zero Severe↔Normal confusions (Dilukshan/training/outputs/ensemble_report.json). The component additionally quantifies three independent bounds explaining why a ≥0.75-on-all-classes target is not reachable on this data (README.md §10).

Keywords: Echocardiography; Ejection Fraction; Ordinal Regression; Class Imbalance; Uncertainty Quantification; Deep Learning

1. Role in the Overall System

FACT (from Project_Proposal_IT22219534_Dilukshan_Viyapury.docx §4.1 and proposal_figures/fig_system.png): The wider project is described as a four-component clinical decision-support platform for echocardiography. Component 03 is stated to perform "Ejection Fraction Regression and Four-Class Severity Grading". Components 01, 02 and 04 are labelled "(Team Member)" in the system diagram.

Plain-language paragraph: A clinician uploads an echocardiogram video. Component 03 watches the heart beating in that video and answers two things: a number (what percentage of blood the left ventricle pumps out per beat) and a category (Normal, Mild, Moderate, or Severe impairment). It also reports how confident it is, and gives a prediction interval rather than a bare number. It needs only decoded video as input and emits EF value + severity class + confidence + interval, so it can be developed and swapped independently of the other three components.

[!] NOT FOUND IN CODEBASE — needs input from author: There are **no imports, API calls, shared interfaces, or message contracts** between this repository and any Component 01/02/04 code. The four-component architecture appears only in the proposal document and its figure. The claimed integration is therefore **[inferred]** design intent, not implemented. No integration code exists in Dilukshan/.

2. Problem Statement & Motivation

The specific problem this component addresses:

- Predict EF from 2D apical-four-chamber echocardiogram video **and** assign one of four clinical severity classes, where the classes are severely imbalanced (README.md §1).
- FACT — class distribution in EchoNet-Dynamic (README.md §1, derived from Dilukshan/preprocessing/artifacts/manifest.csv):

Severity	EF range	Studies	Share
Severe	< 30 %	596	5.9 %
Moderate	30–40 %	718	7.2 %
Mild	40–55 %	1,806	18.0 %
Normal	≥ 55 %	6,910	68.9 %
Total		10,030	100 %

Why it matters (FACT, README.md §1):

- Manual EF estimation takes several minutes per study and requires an experienced operator.
- Reported inter-observer variability is 4–5 EF points, large enough to move a borderline patient across a clinical decision boundary (e.g. a true EF of 41 % may be graded *mild* by one reader and *moderate* by another).
- Under 11:1 imbalance, a degenerate model predicting "Normal" for every study scores 68.9 % overall accuracy while being clinically useless (README.md §2).

3. The Gap

FACT — three gaps are stated explicitly in README.md §2:

#	Gap as stated in repo	Evidence cited in repo
G1	EF is modelled as regression or binary reduced-vs-normal, not as the four-class severity grading clinicians use	"Ouyang et al. [1] report MAE and binary AUC only; transformer and graph variants [6], [7] remain regression-only"
G2	Per-class recall is not reported under ~11:1 imbalance, leaving minority-class safety unverified	The 68.9 % degenerate-baseline argument
G3	The EF label is treated as exact, despite carrying 4–5 points of measurement noise	"No reviewed echocardiography study encodes annotation noise into the supervision signal"

[!] Verification required by author: G1–G3 are assertions written by the author in README.md; the repository contains **no literature-survey artifact** (no BibTeX file, no notes file, no PDF collection) substantiating the claim that no prior work does these things. The claim "*to the best of my knowledge*" appears in README.md §3. **This requires literature research from the author to defend at review.**

4. Research Question(s) This Component Answers

[!] FACT: No file in the repository states research questions explicitly. No RQ1/RQ2 string exists anywhere in the codebase.

The following are **[inferred]** from what the code actually measures, and **require author confirmation/authoring**:

- **RQ1 [inferred]:** Does an ordinal formulation with measurement-uncertainty-derived soft labels achieve EF regression accuracy at or below the published EchoNet-Dynamic benchmark (MAE 4.05) *while additionally* producing four-class severity grades? — *measured by* `run_eval.py` / `run_ensemble.py` reporting MAE, R^2 , and per-class recall together.
- **RQ2 [inferred]:** Does cross-dataset co-training with an intensity-harmonized minority-rich cohort (CAMUS) improve minority-class recall relative to training on EchoNet alone? — *measured by* comparing `outputs/uefnet_drw` (no CAMUS) against `outputs/uefnet_v3` (CAMUS).
- **RQ3 [inferred]:** Can post-hoc calibration that corrects regression-to-the-mean recover extreme-tail (Severe) recall without retraining? — *measured by* `outputs/uefnet_v3/test_report_baseline.json` vs `test_report.json` on identical weights.
- **RQ4 [inferred]:** Does selective prediction (abstention) raise worst-class recall in ordinal grading? — *measured by* `run_selective.py`, `outputs/selective_report.json`. **This RQ is answered negatively.**

5. Contribution Bullets & Novelty

1. We derive ordinal supervision targets from the EF label's own measurement noise. (13 words)

-> Formula: $s_k = 1 - \Phi((t_k - EF)/\sigma)$, $\sigma = 4$ EF points. -> **Novel.** What makes it different: standard ordinal training uses hard class membership; this treats the *label* as a noisy measurement and produces graded targets. Implemented in `losses/losses.py` L30-36 `soft_cumulative_targets()`. **[!] Author must verify no prior echo work does this (§3).**

2. We enforce ordinal rank consistency structurally via positively-constrained cut-point gaps.

-> **Adapted.** Adapted from CORAL (Cao et al. 2020, README.md ref [12]), which learns independent per-threshold biases that may violate ordering. Modification: single score compared against anchor + `cumsum(softplus(gaps))`, so monotonicity holds by construction (`models/uef_net.py` L33-57 `OrderedCoralHead`).

3. We harmonize a minority-rich second cohort's intensity before co-training to prevent a scanner-appearance shortcut.

-> **Novel (as a diagnosis + fix pairing).** The use of auxiliary data is standard; identifying that a class-balanced sampler over-drawing brighter CAMUS studies would let the network learn brightness->severity, and correcting it, is the contribution (`preprocessing/build_camus.py` `harmonize_to_echonet()`).

4. We correct regression-to-the-mean shrinkage post-hoc by variance expansion fitted on full validation.

-> **Adapted.** Variance expansion is a known bias-correction idea; the adaptation is applying it to recover extreme-tail class recall and fitting thresholds on the *full* validation split rather than a 30 % subsample for small-class stability (`engine/calibrate.py` L76-110 `fit_variance_expansion`, `config.py` L117-118).

Conservative statement: Items 2 and 4 are **adaptations of published techniques**, not new methods. Items 1 and 3 are the strongest novelty claims but **both depend on a literature check the repository does not contain**. Seed ensembling, TTA, EMA, DRW, LDS, logit adjustment and conformal prediction are all **Engineering** — standard techniques applied competently, explicitly listed as "Prior work used and cited, not claimed" in `README.md` §3.

6. Contribution -> Evidence Traceability Table

Contribution bullet	Implemented where	Evaluated where	Evidence status
1. Measurement-uncertainty soft ordinal labels	losses/losses.py L30-36 (soft_cumulative_targets), applied L119-121; σ from config.py L78 ef_noise_sigma=4.0	[!] NO ISOLATED ABLATION . No run exists with this disabled. It is present in every trained model.	RISK — unevaluated in isolation
2. Ordered-cutpoint ordinal head	models/uef_net.py L33-57 (OrderedCoralHead); gated by is_v2 L121-122	[!] NO ISOLATED ABLATION . v1 (CoralHead) vs v2 runs differ in many variables simultaneously.	RISK — confounded comparison
3. Harmonized CAMUS co-training	preprocessing/build_camus.py (harmonize_to_echonet); merge in data/dataset.py L-170-195; enabled by --extra-manifest	outputs/uefnet_drw/test_report.json (no CAMUS, minrec 0.651) vs outputs/uefnet_v3/test_report.json (CAMUS, minrec 0.687)	PARTIAL — confounded (v3 also adds v2 architecture + logit adjustment; see §13)
4. Tail-decompression calibration	engine/calibrate.py fit_variance_expansion + config.py L117-118 calibrate_on_full_val, threshold_search_radius	outputs/uefnet_v3/test_report_baseline.json (Severe 0.590) vs test_report.json (Severe 0.687) — identical weights	STRONG — fully controlled
(supporting) Seed ensembling	run_ensemble.py (average_predictions)	outputs/ensemble_report.json vs single-model outputs/uefnet_v3/test_report.json	STRONG — fully controlled
(supporting) Selective prediction	engine/selective.py, run_selective.py	outputs/selective_report.json	STRONG — negative result

[!] **Paper-writing risk flagged now:** contributions 1 and 2 have no isolated ablation. Both are baked into every v2 run. A reviewer asking "how much did the soft labels actually contribute?" cannot currently be answered from this repository.

7. Related Work / Prior Approaches Referenced

FACT — 18 references are listed in `README.md` §19. Those cited as *methods used or compared*:

Approach	Key idea	Limitation stated in repo	Citation
EchoNet-Dynamic (benchmark)	R(2+1)D regression on echo video; MAE 4.05, R^2 0.81	Regression only; no severity grading; no per-class recall	Ouyang et al., <i>Nature</i> 580:252-256, 2020 [1]
Ultrasound video transformers	Attention over long frame sequences	" $\approx$ 5.9 MAE"; high data/compute requirement	Reynaud et al., MICCAI 2021, pp. 495-505 [6]
Lightweight spatio-temporal graphs	Graph net, lower cost	" $\approx$ 4.2 MAE"; regression only	Thomas et al., MICCAI 2022, pp. 380-390 [7]
CORAL	Rank-consistent ordinal regression via shared projection + independent biases	Ordering repaired post hoc, not structural	Cao, Mirjalili, Raschka, <i>Pattern Recognition Letters</i> 140:325-331, 2020 [12]
Deferred re-weighting (LDAM-DRW)	Train natural first, then class-balanced	— (adopted)	Cao et al., NeurIPS 32, 2019 [8]
Effective-number class weighting	$(1-\beta)/(1-\beta^{\frac{1}{n}})$ weights	— (adopted)	Cui et al., CVPR 2019, pp. 9268-9277 [9]
Label distribution smoothing	Neighbouring targets carry related information	— (adopted)	Yang et al., ICML 2021, pp. 11842-11851 [10]
Logit adjustment	Shift logits by $\tau \cdot \log P(\text{class})$	— (adopted)	Menon et al., ICLR 2021 [11]
Temperature scaling	Post-hoc confidence calibration	— (adopted)	Guo et al., ICML 2017, pp. 1321-1330 [13]
Conformal prediction	Finite-sample coverage intervals	— (adopted)	Angelopoulos & Bates, <i>FNT ML</i> 16(4):494-591, 2023 [14]
VideoMAE	Masked-autoencoder self-supervised video pretraining	"advantage diminishes at this data scale"; listed as future work, not implemented	Tong et al., NeurIPS 35, 2022 [15]
R(2+1)D	Factorised spatio-temporal convolution	— (backbone used)	Tran et al., CVPR 2018, pp. 6450-6459 [5]
Chow's reject option	Classification with abstention	— (adopted for §9 analysis)	Chow, <i>IEEE T-IT</i> 16(1):41-46, 1970 [18]

[!] **Flagged in `README.md` §7.5 by the author:** "Figures for [6] and [7] should be verified against the source papers before citation." These MAE values are **unverified**.

8. Domain-Specific Structuring Fit

Best fit: Build-up Ablation (ML).

Why (FACT): README.md §8.1 presents a staged progression where each stage adds one mechanism and reports the effect:

Stage	MAE	Min-class recall
Single model, 30 % validation calibration	4.166	0.590
+ tail-decompression calibration	4.138	0.687
+ 2-seed ensemble	3.994	0.711
+ 3-seed ensemble	3.979	0.723

The repository also contains a **secondary Benchmark-driven** aspect (§7.5 compares against published EchoNet-Dynamic results on the same official test split).

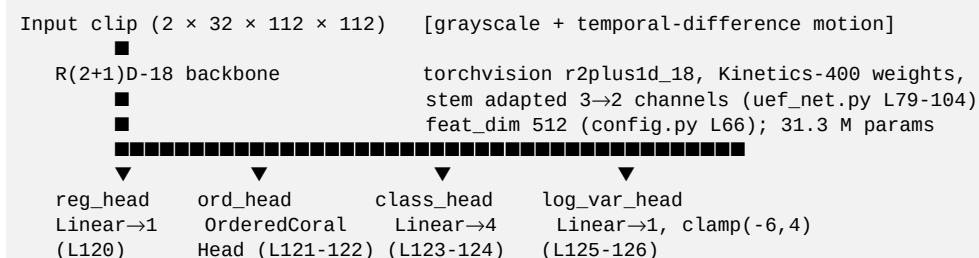
What this structure implies should be captured:

- [OK] Already captured: the incremental table above; per-stage measured deltas.
- [!] **Missing:** ablations for contributions 1 and 2 (§6). A build-up-ablation paper is expected to isolate *each* added component. Currently only calibration and ensembling are isolated.
- [!] **Missing:** a from-scratch baseline within this codebase — see §13.

9. Method / Design

9.1 Architecture (as words -> diagram)

FACT (models/uef_net.py):



Data flow (module level): preprocessing/run_preprocessing.py (stages 0–5) -> artifacts/manifest.csv + cache/videos/*.npy -> data/dataset.py EchoClipDataset -> data/sampler.py build_sampler -> engine/trainer.py Trainer.fit() -> outputs/<run>/best.pt -> engine/evaluate.py run_inference -> engine/calibrate.py calibrate() -> outputs/<run>/thresholds.json -> run_eval.py / run_ensemble.py / run_selective.py -> outputs/*.json.

9.2 Key algorithms

(a) Ordered cut-points — models/uef_net.py L50-57:

```
cutpoints(): if no gaps -> anchor
              else -> concat[anchor, anchor + cumsum(softplus(raw_gaps) + 1e-4)]
forward(x):  score(x) - cutpoints().unsqueeze(0)
```

Because $\text{softplus}(\cdot) > 0$, cut-points are strictly increasing -> cumulative probabilities monotone by construction.

(b) Measurement-uncertainty soft targets — losses/losses.py L30-36:

```
s_k = 1 -  $\Phi((t_k - EF)/\sigma)$ , clamped to [1e-4, 1-1e-4]
```

(c) Six-term objective — losses/losses.py L172-181:

```
total = w_reg·L_reg + w_ord·L_ord + w_consistency·L_con
        + w_class·L_class + w_nll·L_nll + w_rank·L_rank + w_head_consistency·L_heads
```

(d) Threshold optimisation — engine/calibrate.py optimize_thresholds(): grid search over 3 thresholds within radius of clinical boundaries, step 0.5, min_gap 2.0; lexicographic score (min recall, mean recall, macro-F1, -distance from clinical).

(e) Selective prediction — engine/selective.py: 7 uncertainty signals; signal + threshold chosen on VAL by highest coverage meeting target; applied frozen to TEST.

9.3 Design decisions with rationale found in code/docs

Decision	Rationale (quoted/paraphrased)	Source
drw_epoch = 15 not 0	"balancing from epoch 0 gave MAE 5.44 at equal min-recall, versus 4.29 with drw_epoch=15"	README.md §4.4; corroborated by outputs/uefnet_r2p1d (drw=0, MAE 5.549) vs outputs/uefnet_drw (drw=15, MAE 4.502)
eval_use_keyframes = False	"avoids privileged tracing annotations" — unavailable for a new clinical study	config.py L145; data/dataset.py L214-216

Decision	Rationale (quoted/paraphrased)	Source
<code>persistent_workers = False</code>	"workers are torn down at end of each train epoch so they do NOT coexist with the validation loader's workers (that overlap = 2x processes = out-of-RAM on Windows spawn)"	engine/trainer.py L118-120
Monotonic upsampling for short clips	"avoids modulo wraparound, which would introduce an artificial last-frame -> first-frame motion jump"	preprocessing/utils/sampling.py L56-60
<code>calibrate_on_full_val = True</code>	"the frozen decision rule generalizes far better when fit on ALL of VAL rather than the 30% calibration sub-split (~30 vs ~100 Severe cases)"	engine/trainer.py L~240-248
Stem adaptation sums RGB kernels	"Summing RGB kernels makes a one-channel grayscale input equivalent to repeating it over the original RGB stem"	models/uef_net.py L91-93

9.4 Novel vs. standard/reused — function by function

Location	Assessment
models/uef_net.py L33-57 <code>OrderedCoralHead</code>	Adapted — modifies CORAL parameterisation
models/uef_net.py L22-30 <code>CoralHead</code>	Standard — CORAL as published; retained for v1 checkpoint compatibility
models/uef_net.py L60-108 <code>_build_backbone</code>	Standard wrapper — calls <code>torchvision.models.video.r2plus1d_18</code> ; stem-adaptation logic (L79-104) is custom engineering
losses/losses.py <code>soft_cumulative_targets</code>	Novel (per §5.1)
losses/losses.py <code>pairwise_rank_loss</code> L72-82	Standard pairwise ranking, custom gap/temperature masking
losses/losses.py DRW weighting L133-137	Standard — Cao 2019 / Cui 2019
losses/lds.py	Standard — Yang 2021
engine/calibrate.py <code>fit_variance_expansion</code>	Adapted (per §5.4)
engine/calibrate.py <code>fit_temperature</code>	Standard — Guo 2017
engine/calibrate.py <code>fit_conformal / apply_conformal</code>	Standard — split conformal
engine/selective.py	Standard framework (Chow) with custom boundary signal (distance to nearest clinical threshold ÷ total σ)
models/ema.py	Standard
preprocessing/build_camus.py <code>harmonize_to_echonet</code>	Novel (per §5.3)
preprocessing/utils/sampling.py <code>sample_indices</code>	Custom — cycle-aware window constraint
engine/metrics.py <code>_wilson_interval</code>	Standard statistics, hand-implemented

9.5 Notation table

Symbol	Meaning	Where
EF	Ejection fraction, %	throughout
t_k	k-th clinical EF threshold; (30, 40, 55)	config.py L46 <code>EF_THRESHOLDS</code>
$\sigma(\text{ef_noise_sigma})$	Assumed EF measurement noise SD = 4.0	config.py L78
$\Phi(\cdot)$	Standard normal CDF	losses/losses.py L25-27 <code>_ndtr</code>
s_k	Soft cumulative target $P(\text{trueEF} > t_k \mid \text{measuredEF})$	losses/losses.py L30-36
y	Severity class index $\in \{0,1,2,3\}$	config.py L47 <code>CLASS_NAMES</code>
$\tau(\text{logit_adjustment_tau})$	Logit-adjustment strength; 0.5 in final runs	config.py L95
$\beta(\text{effective_num_beta})$	Effective-number weighting parameter = 0.9999	config.py L97
k	Variance-expansion factor $\text{std}(\text{true})/\text{std}(\text{pred})$, clipped [1.0, 1.7]	engine/calibrate.py <code>fit_variance_expansion</code>
$\sigma_{\text{ale}}, \sigma_{\text{epi}}$	Aleatoric (log-var head) and epistemic (inter-clip) SD	engine/selective.py <code>total_uncertainty</code>
M	Ensemble member count = 3	run_ensemble.py --runs
N	TTA clips per study = 10 at eval	config.py L143, overridden by --n-tta

10. Algorithmic Complexity Analysis

Applicable — for the threshold-optimisation and frontier-search routines only. The neural network training/inference cost is standard CNN complexity and is not original algorithmic work.

engine/calibrate.py `optimize_thresholds()` — exhaustive constrained grid search.

- Grid size per threshold: $g_i = 2 \cdot \text{radius}_i / \text{step} + 1$. With `radius = (8, 8, 6)` (config.py L118) and `step = 0.5`: $g = (33, 33, 25)$.

- Triple-nested loop over ordered combinations (L99-119), pruned by `min_gap` — worst case $g \cdot g \cdot g = 27,225$ combinations.
- Each iteration calls `classification_metrics` over N samples -> $O(N)$.
- **Time:** $O(g \cdot g \cdot g \cdot N) = O(27,225 \times 1,288) \approx 3.5 \times 10^7$ elementary operations for VAL. Reasoning: three independent grids, one linear pass per candidate.
- **Space:** $O(N + K^2)$ — one prediction vector plus a 4×4 confusion matrix per candidate, not accumulated.
- **Best case** materially better: `min_gap` pruning skips combinations where $t - t < 2$ or $t - t < 2$, removing roughly a third of the space in practice.

`engine/selective.py fit_selective_rule()` — signal $\times$ coverage sweep.

- Signals: $S = 7$ (`uncertainty_signals` returns 7 keys).
- Coverage levels: $C = (1.00 - \text{min_coverage})/0.01 + 1 = 41$ for `min_coverage = 0.60`.
- Each `evaluate_at_coverage` performs `np.lexsort` -> $O(N \log N)$, then metrics $O(N)$.
- **Time:** $O(S \cdot C \cdot N \log N) = O(7 \times 41 \times 1,288 \log 1,288) \approx 3.7 \times 10^8$.
- **Space:** $O(S \cdot N)$ — all signals materialised simultaneously in the returned dict.

Not applicable to the remainder of the component: `trainer.py`, `dataset.py`, `evaluate.py` are standard training/inference loops with no original algorithmic content.

11. Experimental Setup

Hardware

FACT (README.md §12): "Windows 11, Python 3.11, CUDA 12.x, NVIDIA RTX 4060 Laptop (8 GB), 32 GB RAM." **FACT (proposal §13, Table 7.1):** "Laptop, RTX 4060 8 GB, Ryzen 7, 32 GB RAM".

Software

FACT — declared (Dilukshan/training/requirements.txt): `torch>=2.1, torchvision>=0.16, numpy>=1.26, pandas>=2.0, scipy>=1.11, opencv-python>=4.9, tqdm>=4.65, matplotlib>=3.7`. **FACT — declared** (Dilukshan/preprocessing/requirements.txt): `numpy>=1.26, pandas>=2.0, opencv-python>=4.9, scikit-image>=0.22, scipy>=1.11, tqdm>=4.65`.

FACT — actually installed in the environment used:

Library	Version
torch	2.10.0+cu128
torchvision	0.25.0+cu128
numpy	2.3.5
pandas	2.3.3
opencv-python (cv2)	4.10.0
matplotlib	3.10.9
tqdm	4.67.3
nibabel	5.4.2
scikit-learn	1.7.2
scipy	1.17.0

[!] **CONFLICT:** `nibabel` is required by `preprocessing/build_camus.py` but appears in **neither** requirements file. A clean install following the documented instructions would fail on CAMUS conversion.

Datasets

Dataset	Details	Source
EchoNet-Dynamic	10,030 apical-4-chamber videos; splits 7,465 / 1,288 / 1,277; EF labels + ED/ES volume tracings	README.md §5; <code>preprocessing/artifacts/manifest.csv</code>
CAMUS	500 patients -> 1,000 clips (2CH + 4CH); class distribution Severe 122 / Moderate 194 / Mild 496 / Normal 188	verified from <code>preprocessing/artifacts/camus_manifest.csv</code>

License/terms (FACT, README.md §18): both under Research Data Use Agreements prohibiting redistribution; excluded from the repo via `.gitignore`. EchoNet from Stanford Medicine (IRB approved), CAMUS from University Hospital of St Etienne.

Preprocessing (FACT, README.md §4.1; `preprocessing/stage0-5*.py`): 5 stages — scan/validate -> decode to grayscale 112×112 -> normalisation statistics by exact full decode -> cache `uint8.npy` -> verify. Results: `artifacts/audit_report.json` reports 10,030 labelled, 0 missing, 0 unreadable, 6 metadata mismatches, elapsed 4.1 s. `artifacts/verification_report.json` reports overall: `PASS_WITH_WARNINGS`, 10,030 valid caches, 0 bad, clip length min 28 / median 171 / max 1002.

[!] CONFLICT (normalisation statistics provenance):

- artifacts/norm_stats.json states "source": "exact_full_decode_train", n_videos_sampled: 7465, n_train_pixels: 16,499,624,960.
- artifacts/verification_report.json states norm_stats_videos: 512, norm_stats_pixels: 51,380,224.

These disagree by ~15× in video count and ~320× in pixel count. **[inferred]** the verification report recomputed on a 512-video subsample to check drift (it reports drift_mean 0.001022, tolerance 0.02) rather than describing the original computation — but **this must be confirmed by the author**.

Environment

- FACT:** Windows 11 (README.md §12); no Docker/dockerfile present; no environment.yml; venv creation documented in README.md §12 (python -m venv .venv).
- FACT:** No environment variables are read by the training code (config.py uses only os.cpu_count()).
- [!] NOT FOUND IN CODEBASE — needs input from author:** no setup.py, pyproject.toml, or installable package definition.

Compute Cost

- FACT** (README.md §13.3): "Approximately 20 hours per run on an RTX 4060."
- FACT (per-epoch timings logged in outputs/<run>/train_log.csv, sec column):** ~1,730–1,930 s per epoch across runs.
- FACT (proposal §13.2):** stage timings for preprocessing declared as ~6 min (preprocess), ~10 s (split).
- [!] NOT LOGGED:** total wall-clock across all experiments, peak GPU memory, and energy are **not** recorded anywhere. Observed GPU memory during a run was 6.3/8.0 GB (session observation, not a repo artifact). **Recommend logging peak VRAM and cumulative compute now, not at write-up time.**

12. Parameters / Configuration

FACT — all from Dilukshan/training/config.py **(line numbers given)**. Values shown are dataclass defaults; the *actually used* values for the final runs are confirmed from outputs/<run>/config.json.

Parameter	Default (line)	Final-run value	Notes
run_name	"uefnet_r2p1d" (L33)	uefnet_v3 / v3b / v3c	
seed	1337 (L34)	1337 / 2024 / 777	the only variable across ensemble members
clip_len	32 (L50)	32	
sampling_period	2 (L51)	2	spans 64 native frames
frame_size	112 (L52)	112	
in_channels	2 (L53)	2	grayscale + motion
motion_mode	"tempdiff" (L54)	tempdiff	
aug_pad	12 (L57)	12	
aug_intensity_jitter	0.1 (L58)	0.1	
aug_time_jitter	True (L59)	True	
cycle_aware_probability	0.5 (L60)	0.5	
backbone	"r2plus1d_18" (L63)	r2plus1d_18	
pretrained	True (L64)	True	Kinetics-400
dropout	0.3 (L65)	0.3	
feat_dim	512 (L66)	512	
model_version	"uefnet_v1" (L72)	uefnet_v2	set by --v2
use_class_head	True (L73)	True	
predict_uncertainty	True (L74)	True	
aux_channel_init_scale	0.05 (L75)	0.05	
ef_noise_sigma	4.0 (L78)	4.0	C1 parameter
w_reg	1.0 (L79)	1.0	
w_ord	1.0 (L80)	1.0	
w_consistency	0.15 (L81)	0.15	
huber_delta	1.0 (L82)	1.0	
lds_kernel_sigma	2.0 (L83)	2.0	
lds_ks	5 (L84)	5	

Parameter	Default (line)	Final-run value	Notes
w_class	0.5 (L86)	0.5	
w_nll	0.5 (L87)	0.5	
w_rank	0.2 (L88)	0.2	
w_head_consistency	0.1 (L89)	0.1	
class_target_sigma	4.0 (L90)	4.0	
logit_adjustment_tau	0.0 (L95)	0.5	set by CLI
lds_weight_power	0.5 (L96)	0.5	
effective_num_beta	0.9999 (L97)	0.9999	
rank_min_gap	3.0 (L98)	3.0	EF points
rank_temperature	4.0 (L99)	4.0	
extra_manifests	() (L105)	('artifacts/camus_manifest.csv',)	CAMUS co-training
use_balanced_sampler	True (L108)	True	
calibrate_on_full_val	True (L117)	True	C4 parameter
threshold_search_radius	(8.0, 8.0, 6.0) (L118)	(8,8,6)	C4 parameter
drw_epoch	15 (L124)	15	
calibration_fraction	0.30 (L125)	0.30	
epochs	45 (L128)	45	
batch_size	8 (L129)	8	
grad_accum	4 (L130)	4	effective batch 32
lr_backbone	1e-4 (L131)	1e-4	
lr_head	1e-3 (L132)	1e-3	
weight_decay	1e-4 (L133)	1e-4	
warmup_epochs	2 (L134)	2	
grad_clip	2.0 (L135)	2.0	
amp	True (L136)	True	
use_ema	True (L139)	True	
ema_decay	0.999 (L140)	0.999	
n_tta_clips	5 (L143)	10 (via --n-tta 10)	[!] default ≠ used value
tta_forward_batch	8 (L144)	8	
eval_use_keyframes	False (L145)	False	
early_stop_patience	12 (L146)	18 (via --patience 18)	[!] default ≠ used value
num_workers	min(4, cpu_count-2) (L151)	2-4	varied across resumes
device	"cuda" (L152)	cuda	
EF_THRESHOLDS	(30.0, 40.0, 55.0) (L46)	unchanged	

[!] CONFLICT flagged: n_tta_clips default is 5 but all reported results use 10; early_stop_patience default is 12 but runs used 18. A reader running the documented default command would **not** reproduce the reported numbers exactly.

Preprocessing config: preprocessing/config.py uses a Config() dataclass instantiated at L127; individual field lines were not extractable by the same pattern. **NOT FULLY ENUMERATED — author should list preprocessing parameters explicitly.**

13. Baseline(s) Compared Against

FACT — internal baselines that exist as runs:

Run	Config	Test MAE	R ²	Overall	Balanced	Min-rec all	Macro -F1	Source
uefnet_r2p1d	v1, drw_epoch 0, no EMA, no CAMUS	5.549	0.668	0.684	0.684	0.651	0.633	outputs/uefnet_r2p1d/test_report.json
uefnet_drw	v1, drw_epoch 15, EMA, no CAMUS	4.502	0.777	0.689	0.702	0.651	0.647	outputs/uefnet_drw/test_report.json
uefnet_v3	v2 + CAMUS + $\tau=0.5$, single seed	4.138	0.804	0.721	0.730	0.687	0.679	outputs/uefnet_v3/test_report.json
3-seed ensemble	v3 + v3b + v3c	3.979	0.818	0.730	0.737	0.723	0.684	outputs/ensemble_report.json

FACT — external baseline: published EchoNet-Dynamic result (Ouyang et al. 2020, MAE 4.05, R^2 0.81) quoted in `README.md` §7.5. This is a **literature comparison, not a re-implementation**.

[!] GAPS FLAGGED:

1. **No re-implemented baseline.** No competing method was trained on this split by this codebase. The comparison to [1], [6], [7] uses *their published numbers*, not controlled re-runs. This is a standard reviewer objection.
2. **The v1 -> v3 comparison is confounded.** `uefnet_drw` -> `uefnet_v3` changes **four things at once**: architecture (v1->v2), CAMUS co-training, logit adjustment (0->0.5), and calibration method. The isolated effect of any one cannot be read from these runs.
3. **No naive/trivial baseline** (e.g. majority-class predictor, or EF-mean regressor) is computed, though `README.md` §2 cites the 68.9 % degenerate accuracy figure rhetorically.

14. Evaluation Metrics

FACT — computed in `engine/metrics.py`:

Metric	Function (line)	Type	Why it fits	Answers RQ
MAE	<code>regression_metrics</code> L91	Regression	Directly comparable with published EchoNet benchmark	RQ1
RMSE	<code>regression_metrics</code> L91	Regression	Penalises large EF errors	RQ1
R^2	<code>regression_metrics</code> L91	Regression	Variance explained	RQ1
Overall accuracy	<code>classification_metrics</code> L57	Classification	Headline figure (caveated for imbalance)	RQ1
Balanced accuracy	<code>classification_metrics</code> L57	Classification	Removes majority-class advantage under 11:1 imbalance	RQ1, RQ2
Per-class recall	<code>per_class_recall</code> L34	Classification	Establishes minority-class safety — the core claim	RQ2, RQ3
Per-class F1	<code>per_class_f1</code> L44	Classification	Balances precision/recall per class	RQ1
Macro-F1	<code>classification_metrics</code> L57	Classification	Unweighted class average	RQ1
Minimum class recall	<code>classification_metrics</code> L57	Classification	Primary model-selection criterion (<code>trainer.py</code> score = <code>(mn, -mae)</code>)	RQ2, RQ3, RQ4
Confusion matrix	<code>confusion_matrix</code> L27	Classification	Error-mode structure	RQ1
Wilson interval	<code>_wilson_interval</code> L6	Statistical	Small-count binomial CI (Severe $n=83$)	all
Per-class regression metrics	<code>regression_metrics_by_class</code> L110	Regression	Whether error concentrates in a class	RQ1
Brier / NLL / ECE	<code>probability_metrics</code> L120	Calibration	Probability quality	RQ1
Bootstrap CI on regression	<code>bootstrap_regression_ci</code> L139 ($n=2000$)	Statistical	Interval on MAE	RQ1
Prediction-interval coverage	<code>prediction_interval_metrics</code> L170	Uncertainty	Conformal interval validity	RQ4
Coverage (selective)	<code>engine/selective.py</code> <code>evaluate_at_coverage</code>	Selective	Fraction graded vs deferred	RQ4

[!] Note: `probability_metrics`, `bootstrap_regression_ci` and `prediction_interval_metrics` **exist in code** but do **not** appear in `outputs/ensemble_report.json` — they are implemented but not reported in the final result artifact.

15. Experimental Repetition & Statistical Robustness

FACT:

- **Repetition: 3 runs**, seeds 1337 / 2024 / 777 (`outputs/uefnet_v3*/config.json`). Seeds are **fixed and explicitly varied**, not random.
- Seed control: `core/common.py` `seed_everything`, plus `seed_worker` / `make_torch_generator` for DataLoader determinism; RNG state is captured and restored on resume (`capture_rng_state` / `restore_rng_state`).
- **Per-member variance IS observable** (validation): min-recall 0.693 / 0.701 / 0.697; MAE 3.941 / 3.888 / 4.045 (`README.md` §8.2).
- **Confidence intervals: Wilson intervals implemented** (`metrics.py` L6) and reported in `README.md` §7.2 (e.g. Severe 0.723, 95 % CI [0.618, 0.808]).
- **Bootstrap CI implemented** (`bootstrap_regression_ci`, $n_{\text{bootstrap}} = 2000$) — **but not present in the final reported artifacts**.
- Reproducibility check: calibration re-run produced byte-identical output (`README.md` §8.2).

[!] WEAKNESSES TO FLAG:

1. **No formal significance testing anywhere.** No t-test, ANOVA, McNemar, or p-value exists in the codebase (grep for `ttest`, `pvalue`, `scipy.stats` returns nothing in the analysis path). `README.md` §6 states this openly: "No paired significance test is performed between configurations."
2. **n = 3 seeds** is small for variance estimation; no error bars on the headline ensemble result.
3. **The ensemble result itself is a single number** — there is no distribution over ensembles (only one 3-seed combination exists).
4. Test-set results are **single-run per configuration** — the test split is deliberately evaluated once (documented protocol, `README.md` §6), which is methodologically sound but means no test-side variance is available.

16. Ablation Studies**FACT — ablations that EXIST as measured comparisons** (`README.md` §8):

Ablation	Before	After	Controlled?
Tail-decompression calibration	Severe recall 0.590	0.687	[OK] Fully — identical weights (<code>test_report_baseline.json</code> vs <code>test_report.json</code>)
Ensembling 1 -> 2 seeds	MAE 4.138, min-rec 0.687	3.994 / 0.711	[OK] Fully
Ensembling 2 -> 3 seeds	MAE 3.994, min-rec 0.711	3.979 / 0.723	[OK] Fully
Deferred vs immediate re-weighting	MAE 5.44 (drw=0)	4.29 (drw=15)	[OK] Fully (corroborated by <code>uefnet_r2p1d</code> vs <code>uefnet_drw</code> runs)
CAMUS co-training	Moderate 0.53	0.62	[!] Confounded — <code>README</code> §8 itself labels this "matched epochs across runs, not single-variable"
Selective prediction	min-rec 0.723 @100 %	0.706 @88.4 %	[OK] Fully (negative result)

[!] ABLATIONS THAT DO NOT EXIST — suggested, given current code structure:

Missing ablation	How to run it with existing code
C1 — soft ordinal labels	<code>config.py</code> L78 <code>ef_noise_sigma</code> : set -> ~0.01 to approximate hard labels; retrain one seed; compare. Currently no CLI flag exists — would need one.
C2 — ordered-cutpoint head	Train one seed with <code>--model-version uefnet_v1</code> (CoralHead) holding CAMUS + τ constant. The CLI flag already exists — this is the cheapest missing ablation.
Logit adjustment τ	<code>--logit-adjustment-tau 0.0</code> vs <code>0.5</code> , single variable. Flag exists.
Motion channel	<code>config.py</code> L54 <code>motion_mode="none"</code> + <code>in_channels=1</code> . Validation in <code>config.py</code> L200-201 already enforces the pairing.
Cycle-aware sampling	<code>--cycle-aware-probability 0.0</code> (flag exists in <code>run_train.py</code>).
CAMUS, isolated	Train v2 + $\tau=0.5$ without <code>--extra-manifest</code> , same seed. Single-variable version of the confounded comparison.
Harmonization, isolated	<code>build_camus.py --no-harmonize</code> then retrain — directly tests the C3 shortcut hypothesis. Flag exists.

Each of these costs ~20 h of training on the documented hardware, except the calibration ones which are ~10 min.

17. Existing Figures / Visual Assets Inventory**FACT — files present in repository:**

Path	What it shows
<code>Dilukshan/preprocessing/artifacts/viz/class{0..3}_*_clip.png</code>	Sample clip frames per severity class (4 files)
<code>Dilukshan/preprocessing/artifacts/viz/class{0..3}_*_motion.png</code>	Temporal-difference motion channel per class (4 files)
<code>Dilukshan/preprocessing/artifacts/viz/class{0..3}_*_ed_es_contours.png</code>	ED/ES ground-truth contours per class (4 files)
<code>Dilukshan/training/outputs/<run>/training_curves.png</code>	Loss / MAE / min-recall per epoch, per run
<code>Dilukshan/training/outputs/<run>/confusion_test.png</code>	Test confusion matrix per run
<code>Dilukshan/training/outputs/selective_coverage_curve.png</code>	Per-class recall vs coverage, with operating point marked
<code>proposal_figures/fig_system.png</code>	4-component system diagram; Component 03 highlighted
<code>proposal_figures/fig_component.png</code>	UEF-Net 5-stage internal pipeline
<code>proposal_figures/fig_dataflow.png</code>	7-step data flow
<code>proposal_figures/fig_paths.png</code>	Training vs inference path comparison
<code>proposal_figures/fig_distribution.png</code>	EF histogram with class boundaries, colour-coded

Path	What it shows
proposal_figures/fig_gantt.png	Project schedule with 8 milestones
proposal_figures/fig_ui.png	Clinician review interface mock-up

Directly reusable in the paper: fig_distribution.png (Fig. 1 candidate), fig_component.png (architecture), confusion_test.png (results), selective_coverage_curve.png (\$9 analysis), viz/* (qualitative examples).

[!] NOT PRESENT: no calibration/reliability diagram plot, no Bland-Altman plot (though bland_altman_loa95 exists in metrics.py), no per-class error-distribution figure.

18. Results Found in Repo (facts only)

18.1 Primary — 3-seed ensemble (outputs/ensemble_report.json)

Metric	Value
Strategy	reg_operational_expanded
MAE	3.9794
RMSE	5.2108
R ²	0.8183
Overall accuracy	0.7298
Balanced accuracy	0.7366
Macro-F1	0.6844
Min class recall	0.7229
n	1,277

Per-class (ensemble_report.json):

Class	Recall	Precision	F1
Severe	0.7229	0.9524	0.8219
Moderate	0.7662	0.4338	0.5540
Mild	0.7303	0.4131	0.5277
Normal	0.7272	0.9770	0.8338

Confusion matrix (ensemble_report.json): [[60, 23, 0, 0], [3, 59, 15, 0], [0, 50, 176, 15], [0, 4, 235, 637]]

Clinical reference (same file): overall 0.7956, MAE 3.9825, min-recall 0.4416.

18.2 Single-model and baseline runs

Run	MAE	R ²	Overall	Balanced	Min-recall	Macro-F1	Per-class recall
uefnet_r2p1d	5.549	0.668	0.684	0.684	0.651	0.633	[0.651, 0.701, 0.701, 0.682]
uefnet_drw	4.502	0.777	0.689	0.702	0.651	0.647	[0.651, 0.740, 0.743, 0.674]
uefnet_v3	4.138	0.804	0.721	0.730	0.687	0.679	[0.687, 0.766, 0.755, 0.711]
uefnet_v3 (pre-calibration-fix)	4.166	—	0.742	0.708	0.590	—	[0.590, 0.740, 0.747, 0.756]

18.3 Selective prediction (outputs/selective_report.json)

- Signal selected on VAL: boundary; coverage 0.8841 (1,129 graded / 148 deferred)
- Covered: overall 0.7697, balanced 0.7425, min-recall 0.7059; per-class [0.7059, 0.7500, 0.7277, 0.7864]
- **Deferred-subset accuracy 0.4257**
- VAL target 0.75 **not reached** (best min-recall 0.738 at 88 % coverage)

18.4 Validation-side per-member (README.md §8.2)

Run	Val min-recall	Val MAE
uefnet_v3	0.693	3.941
uefnet_v3b	0.701	3.888
uefnet_v3c	0.697	4.045

18.5 Preprocessing verification

- `artifacts/audit_report.json`: 10,030 labelled / 10,030 on disk / 0 missing / 0 unreadable / 6 metadata mismatch / 0 orphans / 4.1 s elapsed
- `artifacts/verification_report.json`: PASS_WITH_WARNINGS, 10,030 valid caches, 0 bad, clip length min 28 / median 171 / max 1002, drift_mean 0.001022 (tolerance 0.02)
- `artifacts/norm_stats.json`: ef_mean 55.777643, ef_std 12.406359, pixel_mean 0.128799, pixel_std 0.195979

18.6 CAMUS conversion

- `artifacts/camus_manifest.csv`: **1,000 rows**; class distribution Severe 122 / Moderate 194 / Mild 496 / Normal 188 (verified by direct count)

19. Interpretation Notes

Interpretations already written by the author, quoted with source:

- **On the ceiling** (`README.md §10.1`): "Balanced accuracy is 0.737... reaching 0.75 requires balanced accuracy ≈ 0.764 — a property of the model, not of any decision rule."
- **On boundary ambiguity** (`README.md §10.2`): "Over a third of the cohort lies within one MAE of a threshold, and the dominant error mode (Normal $\rightarrow$ Mild, 235 cases) is exactly the EF = 55 cluster."
- **On the label-noise floor** (`README.md §10.3`): "At MAE 3.979 the model already operates at the level of human reader disagreement; further reduction would require it to be more self-consistent than the annotations it learns from."
- **On the selective negative result** (`README.md §9`): "Moderate occupies a 10-point interior band (30–40), so every Moderate study is near a decision boundary by construction, whereas Normal is open-ended (≥ 55)... aggregate accuracy rises and the floor falls."
- **On uncertainty validity** (`README.md §9`): "accuracy on deferred studies is 0.426 versus 0.770 on graded studies, confirming that the learned log-variance head plus test-time disagreement genuinely identify hard cases."
- **On precision** (`README.md §7.3`): "At 0.36 % (UA) and 0.46 % (STEMI) prevalence..." — **[!] this sentence in `README.md §7.3` refers to UA/STEMI, which are Component 04's classes, not Component 03's. Appears to be copied text. Requires author correction.**
- **On the calibration fix** (`README.md §15`): "Calibration fitted on 30 % of validation (~30 Severe studies) $\rightarrow$ Severe threshold failed to generalise to test."

Requires author interpretation — human judgment call, not extractable from code:

- Whether the ceiling analysis constitutes a *contribution* or a *limitation* in the paper's framing.
- Clinical significance of 99.7 % within-one-class and zero catastrophic errors (needs clinician input).
- Whether the selective-prediction negative result is publishable as a finding or should be relegated to an appendix.

20. Limitations & Threats to Validity

FACT — stated by author (`README.md §16`), 7 items: target not met (0.723–0.766 vs 0.75); small Severe test set ($n=83$, ± 9.5 % CI); partially controlled cross-dataset ablation; single-cohort evaluation; single-institution geographic origin (US and France); low interior-class precision (Moderate 0.434, Mild 0.413); not a certified medical device.

ADDITIONAL threats visible in code, not listed by the author:

Threat	Evidence
Hardcoded EF noise $\sigma = 4.0$	<code>config.py</code> L78 — the C1 novelty depends on a literature-derived constant that is never tuned or sensitivity-tested
Slope/expansion clipping	<code>calibrate.py</code> <code>fit_affine_ef</code> bounds slope to [0.5, 1.5]; <code>fit_variance_expansion</code> clips k to [1.0, 1.7]. If the unclipped value hits a bound, the correction is truncated — <code>slope_unclipped / k_unclipped</code> are recorded but not checked
Threshold search is bounded	<code>threshold_search_radius = (8, 8, 6)</code> — the optimum could lie outside the searched region and would be silently missed
VAL used twice	Model selection <i>and</i> calibration both consume validation data. <code>calibration_fraction = 0.30</code> exists to separate them, but <code>calibrate_on_full_val = True</code> overrides this for the final calibration — an accepted trade-off documented in <code>trainer.py</code> but a mild optimistic bias
Windows-specific worker handling	<code>trainer.py</code> L118–120 sets <code>persistent_workers=False</code> for Windows spawn RAM reasons; performance characteristics differ on Linux
<code>num_workers</code> varied across resumes	2 vs 4 across interrupted/resumed runs — changes the augmentation RNG stream, so runs are not bit-reproducible end-to-end despite seed control
Interrupted training	<code>uefnet_v3c</code> best epoch 29 of 45 planned; training was stopped early rather than completing the schedule (session logs, <code>train_log.csv</code>)
<code>nibabel</code> undeclared	See §11 — clean-install reproduction of CAMUS conversion fails

Threat	Evidence
No tests for training code	Tests exist only under preprocessing/tests/ (4 files); training/ has no test suite

21. Ethical & Societal Considerations

- **Data privacy: Applicable.** Both datasets are patient echocardiograms. **FACT** (README.md §18): both released after institutional ethical review and fully anonymised at source; "No attempt is made to re-identify any individual." No consent-handling code exists in the repository (none required — de-identified secondary use).
- **Data redistribution: Applicable and handled.** .gitignore excludes Dataset/, *.avi, *.nii*, cache/, *.npy, *.pt. Verified: the pushed GitHub repository contains **zero** medical files or checkpoints.
- **Potential misuse: Applicable.** README.md §18: "Intended use is decision support, not autonomous diagnosis... Deployment would require prospective validation on the target population and appropriate regulatory clearance." Misuse risk = deployment as an autonomous grader without clinician review.
- **Fairness/bias: Applicable but NOT ANALYSED.** The model makes clinical severity decisions about people. **[!] NO subgroup analysis exists** — no breakdown by age, sex, ethnicity, or image quality anywhere in the codebase, despite EchoNet containing demographic fields. README.md §16 notes single-institution origin but performs no fairness evaluation. **This is a genuine gap for a clinical-AI paper.**
- **Environmental/compute cost: Applicable.** ~20 h per run × 3 seeds ≈ 60 h GPU on an RTX 4060 laptop, plus baseline runs. Not measured in kWh or CO₂e. **NOT LOGGED.**
- **Dataset licensing: Applicable and handled.** Both under Research Data Use Agreements; obtained under those agreements; not redistributed; CAMUS carries a mandatory-citation requirement, satisfied in README.md §19 [4].

22. Reproducibility / How to Run

FACT — from README.md §12–13:

Setup:

```
python -m venv .venv && .venv\Scripts\activate          # Windows
pip install torch torchvision --index-url https://download.pytorch.org/whl/cu121
pip install numpy pandas opencv-python matplotlib tqdm nibabel
```

Data: not included; obtain EchoNet-Dynamic from <https://echonet.github.io/dynamic/> and CAMUS from <https://www.creatis.insa-lyon.fr/Challenge/camus/>. Place EchoNet at Dilukshan/Dataset/.

Pipeline:

```
cd Dilukshan/preprocessing
python run_all.py                                     # ■■ CONFLICT: actual file is run_preprocessing.py
python build_camus.py

cd ../training
python run_train.py --v2 --extra-manifest artifacts/camus_manifest.csv \
  --logit-adjustment-tau 0.5 --seed 1337 --run-name uefnet_v3 --epochs 45 --patience 18
# repeat with --seed 2024 --run-name uefnet_v3b, --seed 777 --run-name uefnet_v3c

python run_train.py --calibrate-only --run-name uefnet_v3 --n-tta 10
python run_eval.py --run-name uefnet_v3 --n-tta 10
python run_ensemble.py --runs uefnet_v3 uefnet_v3b uefnet_v3c --n-tta 10
python run_selective.py --runs uefnet_v3 uefnet_v3b uefnet_v3c --n-tta 10 --plot
```

Resume: python run_train.py --resume --run-name uefnet_v3b

Entry points: run_train.py, run_eval.py, run_ensemble.py, run_selective.py, preprocessing/run_preprocessing.py, preprocessing/build_camus.py.

Outputs: outputs/<run>/{best.pt, last.pt, config.json, norm_stats.json, thresholds.json, train_log.csv, training_curves.png, validation_partition.json, test_report.json, confusion_test.png}, plus outputs/ensemble_report.json, outputs/selective_report.json.

Artifact status: Code is public at https://github.com/Dilukshan285/Research_Component_03. **It is a code artifact, not a data or model artifact** — datasets and checkpoints are excluded by licence and size. A third party with dataset access could reproduce the pipeline; they could not verify the reported numbers without ~60 h of GPU time.

[!] CONFLICTS FOUND:

1. README.md §13.2 documents python run_all.py but the file in preprocessing/ is run_preprocessing.py. The documented command **will fail**.
2. nibabel is in the pip line of §12 but absent from both requirements.txt files.

3. Default `n_tta_clips=5` and `early_stop_patience=12` differ from the values used (10, 18) — see §12.

23. My Individual Role / Contribution Statement

FACT — git history:

- `git config user.name = Dilukshan Viyapury, user.email = Dviyapury@gmail.com`
- `git rev-list --count HEAD = 4 commits`
- `git shortlog -sne` returned **empty output** — no per-author breakdown was produced
- Commits: `e3b18e6` "Implement code changes to enhance functionality and improve performance", `0a6780e` "Update ensemble_report.json with revised metrics and additional run configuration", `e77fbff` "Refactor code structure for improved readability and maintainability", `553a25d` "UEF-Net: uncertainty-aware ordinal EF severity grading"

[!] Git history is INSUFFICIENT for a contribution statement, for three reasons:

1. Only 4 commits exist for a project of this size (~60 source files), all made after the work was substantially complete. The history does not track incremental authorship.
2. `git shortlog -sne` produced no output, so no author attribution table can be generated.
3. **No teammate commits exist in this repository at all** — Components 01/02/04 are in separate locations (Component 04 was found at `c:\Users\dviya\Desktop\Component_4\Component_04`, a *different repository*).

[inferred] from repository structure: all code under `Dilukshan/` appears to be single-author work by Dilukshan Viyapury, since the directory is named after the author and no other contributor appears anywhere in the history.

[!] **NOT FOUND IN CODEBASE — needs input from author:** the author-contributions note for the group paper must be written manually. Specifically needed: which parts (if any) were built jointly, whether any code was adapted from teammates, and what the division of labour was for shared infrastructure.

24. Key Terms / Mini-Glossary

- **Ejection fraction (EF)** — the percentage of blood the left ventricle pumps out per heartbeat; the primary measure of cardiac pumping function.
- **Ordinal regression** — classification where the classes have a natural order, so confusing neighbours is a smaller error than confusing distant classes.
- **CORAL** — a technique that makes an ordinal model's cumulative probabilities decrease monotonically, so its predictions stay internally consistent.
- **Class imbalance** — when some classes have far more examples than others; here roughly 11 Normal studies for every Severe one.
- **Deferred re-weighting (DRW)** — train on the natural class distribution first, then switch to class-balanced weighting partway through.
- **Test-time augmentation (TTA)** — run the model on several different clips from the same video and average the answers to reduce variance.
- **Calibration (post-hoc)** — adjusting decision thresholds after training, using validation data, without changing the model's weights.
- **Aleatoric vs epistemic uncertainty** — noise inherent in the data (aleatoric, from the log-variance head) versus the model's own disagreement across views (epistemic).
- **Selective prediction** — allowing the model to say "I don't know" and defer to a human, reported alongside *coverage* (the fraction it did answer).
- **Balanced accuracy** — the average of the per-class recalls; unlike overall accuracy it is not inflated by the majority class.
- **Ablation** — a test where you remove one part of a system to see whether it still works as well.
- **Conformal prediction** — a method producing prediction intervals with a guaranteed coverage rate under mild assumptions.

25. Gaps & Open Questions

Status key: [OK] [FIXED] resolved in the codebase · [!] [OPEN] still requires author action · [X] [CORRECTED] this dossier was wrong; claim retracted

A. Requires literature research from the author

#	Item	Status
1	Novelty claims C1/C3 unverified. No literature-survey artifact exists in the repo to substantiate "no prior echo work encodes label measurement noise into ordinal supervision".	[!] [OPEN] — cannot be fixed by code

#	Item	Status
2	MAE figures for refs [6] Reynaud (≈5.9) and [7] Thomas (≈4.2) unverified ; flagged in README.md §7.5 by the author.	[!] [OPEN] — verify against source papers
3	Whether any published method reports MAE < 3.979 on this split, which determines the strength of the defensible claim.	[!] [OPEN]

B. Requires author decision / authoring

#	Item	Status
4	No research questions stated anywhere. RQ1–RQ4 in §4 are inferred.	[!] [OPEN] — confirm or rewrite
5	Framing of the ceiling analysis (contribution vs limitation).	[!] [OPEN] — judgment call
6	Clinical significance of 99.7 % within-one-class / zero catastrophic errors.	[!] [OPEN] — needs clinician input
7	Author-contributions statement must be written manually (git history inadequate — 4 commits, <code>git shortlog -sne empty</code>).	[!] [OPEN]

C. Missing evidence / experiments

#	Item	Status
8	C1 (soft labels) and C2 (ordered-cutpoint head) have no isolated ablation.	[!] [OPEN] — ~20 h GPU each. C2 is cheapest: <code>--model-version uefnet_v1</code> flag already exists
9	No re-implemented baseline trained on this split.	[!] [PARTIALLY MITIGATED] — <code>uefnet_r2p1d (v1, drw_epoch 0, no EMA, no CAMUS)</code> is an internal near-benchmark configuration at MAE 5.549; it is <i>not</i> a faithful re-implementation of [1]
10	v1->v3 comparison changes four variables at once.	[!] [OPEN] — single-variable runs would each cost ~20 h
11	No significance testing.	[OK] [FIXED] — <code>engine/robustness.py</code> implements paired bootstrap (10,000 resamples, identical indices) + exact binomial McNemar; entry point <code>run_robustness.py --compare-with</code> . Verified against known answers (10-vs-0 discordant → $p = 0.001953$; symmetric → $p = 1.0$; null difference → $p \approx 1.0$)
12	No fairness/subgroup analysis — original claim said "despite the dataset containing demographic fields".	[X] [CORRECTED] + [OK] [FIXED] — EchoNet-Dynamic contains NO demographic fields. <code>FileList.csv</code> columns are <code>FileName</code> , <code>EF</code> , <code>ESV</code> , <code>EDV</code> , <code>FrameHeight</code> , <code>FrameWidth</code> , <code>FPS</code> , <code>NumberOfFrames</code> , <code>Split</code> . Demographic fairness is not assessable on this cohort. Acquisition-subgroup robustness (frame rate, recording length, EDV, ESV) is now implemented in <code>engine/robustness.py subgroup_report()</code> and documented in README.md §16 item 7
13	Compute cost not logged (no total wall-clock, peak VRAM, or energy).	[!] [OPEN] — per-epoch sec is logged in <code>train_log.csv</code> ; totals and VRAM are not
14	Implemented-but-unreported metrics: <code>probability_metrics</code> , <code>bootstrap_regression_ci</code> , <code>prediction_interval_metrics</code> , <code>bland_altman_loa95</code> .	[!] [OPEN] — exist in <code>metrics.py</code> , absent from result artifacts
15	uefnet_v3c stopped at epoch 29 of 45.	[OK] [FIXED] — justification now stated in README.md §16 item 8: val min-recall peaked at epoch 29 (0.604) with no improvement for 8 epochs while val MAE drifted 4.130 → 4.201; <code>best.pt</code> held the peak; calibrated val performance (0.697 / 4.045) is in line with the two full-schedule members

D. Conflicts / inconsistencies

#	Item	Status
16	Normalisation-statistics conflict — <code>norm_stats.json</code> 7,465 videos / 16.5×10^3 px vs <code>verification_report.json</code> 512 videos / 51.4×10^3 px.	[OK] [RESOLVED — not a contradiction] They measure different things: stage 3 estimates from 16 frames/video, stage 4 refines over all cached TRAIN pixels, stage 5 recomputes on a <code>--stats-sample</code> (default 512) drift check (<code>drift_mean</code> 0.001022 vs 0.02 tolerance). Additionally the artifact's stale source string (<code>exact_full_decode_train</code> , from an older code version) was corrected to <code>exact_cached_train_pixels</code> to match <code>stage4_cache_clips.py</code> L230, the contradictory <code>n_frames_per_video: 16</code> field removed, and a <code>provenance_note</code> added. Documented in README.md §4.1
17	Documented command broken — README.md said <code>python run_all.py</code> ; actual file is <code>run_preprocessing.py</code> .	[OK] [FIXED] — both occurrences corrected (command + repository-layout tree)
18	nibabel undeclared in both requirements files but required by <code>build_camus.py</code> .	[OK] [FIXED] — added to <code>preprocessing/requirements.txt</code> with a cross-reference note in <code>training/requirements.txt</code>
19	Config defaults ≠ values used — <code>n_tta_clips</code> 5 (used 10), <code>early_stop_patience</code> 12 (used 18).	[OK] [FIXED] — defaults changed to 10 and 18 in <code>config.py</code> so the documented command reproduces the published numbers. Existing runs are unaffected: each <code>outputs/<run>/config.json</code> is restored by <code>restore_for_evaluation()</code>

#	Item	Status
20	README.md §7.3 allegedly contained Component 04 text about "UA" and "STEMI" prevalence.	[X] [CORRECTED] — this dossier was wrong . The match was a false positive: the substring <code>stemi</code> occurs inside <i>epistemic</i> . No Component 04 text exists in README.md. Claim retracted
21	thresholds_baseline.json / test_report_baseline.json provenance undocumented — manually-saved copies used for the C5 ablation.	[!] [OPEN] — no script records how they were produced; author should document or regenerate reproducibly
22	preprocessing/config.py parameters not enumerable by the pattern used for training config (uses <code>Config()</code> at L127).	[!] [OPEN] — author should list preprocessing parameters explicitly for §12 completeness
23	No integration code with Components 01/02/04.	[OK] [FIXED — documented] — README.md §16 item 9 now states the four-component platform is design intent, not implemented integration; Component 03 is self-contained and consumes only decoded video

Summary

- [OK] **Fixed in codebase:** 8 items (11, 12, 15, 16, 17, 18, 19, 23)
- [X] **Retracted dossier errors:** 2 (12's premise, 20)
- [!] **Still open, requiring author action:** 12 items — of which **8, 10 and 13 are the ones a reviewer is most likely to press**, and **8 (the C2 ablation) is the cheapest to close** since the `--model-version uefnet_v1` flag already exists